

GUARDIAN EYE

Real-Time Abnormal Activity Detection in Video Surveillance Using Computer Vision · June 2026

0	8–12 frames	2.6×	3	No GPU
False Positives (Normal)	Detection Latency	Signal Discriminability	Test Configurations	Hardware Required

1. PROJECT OVERVIEW

Guardian Eye is a lightweight, intelligent surveillance system that automatically detects abnormal human activities in real-time video streams — **without requiring deep learning frameworks or GPU hardware**. Built on classical computer vision techniques (**MOG2 background subtraction** + **Farneback Dense Optical Flow**), it runs on commodity laptops, making it immediately deployable in educational institutions, public spaces, and security installations.

Component	Detail
Technology	Python 3.x · OpenCV 4.x · NumPy · Pygame 2.x
Core Methods	MOG2 Background Subtraction · Farneback Dense Optical Flow
Alert Output	Synthesised 880 Hz audio alarm · On-screen red warning banner
Hardware	Intel Core i5 / 8 GB RAM — no GPU required
Status	Functional proof-of-concept validated across 3 test configurations

2. HOW IT WORKS — 3-MODULE PIPELINE

① Motion Detector	MOG2 background subtraction isolates foreground movement from a dynamically updating scene model. Morphological closing and dilation remove noise and merge nearby blobs into person-scale regions.
② Activity Classifier	3-rule engine using Farneback optical flow: sudden spike detection (Rule 1), sustained high-motion check (Rule 2), and fallen-person aspect-ratio analysis (Rule 3). A temporal accumulator with hysteresis prevents false triggers.
③ Alert System	Confirmed abnormal events trigger an 880 Hz synthesised audio alarm via Pygame and a full-width red on-screen warning banner — dispatched asynchronously on a daemon thread with a 10-second cooldown.

3. KEY RESULTS

- ◆ **Normal Video:** Zero false positives — the abnormal counter never exceeded 3 during any normal footage run.
- ◆ **Fighting Video:** ABNORMAL state confirmed within **8–12 frames (~267 ms)** of fight onset; peak motion scores exceeded 1,000,000 (vs threshold of 350,000).
- ◆ **Live Webcam:** Simulated aggressive gestures reliably triggered audio + visual alerts with no dropped frames.
- ◆ **Calibration Margin:** Mean motion score 156,648 (normal) vs 413,663 (fighting) — a clear **2.6× discriminability ratio**.

4. SYSTEM OUTPUT — LIVE DETECTION SCREENSHOTS

STATUS: NORMAL

Ordinary pedestrian activity detected. Green bounding boxes drawn around persons. No alert triggered — abnormal counter remained below threshold throughout.

— LIVE DETECTION SCREENSHOTS

NORMAL

detected. Green bounding boxes drawn around persons. No alert triggered throughout.

System

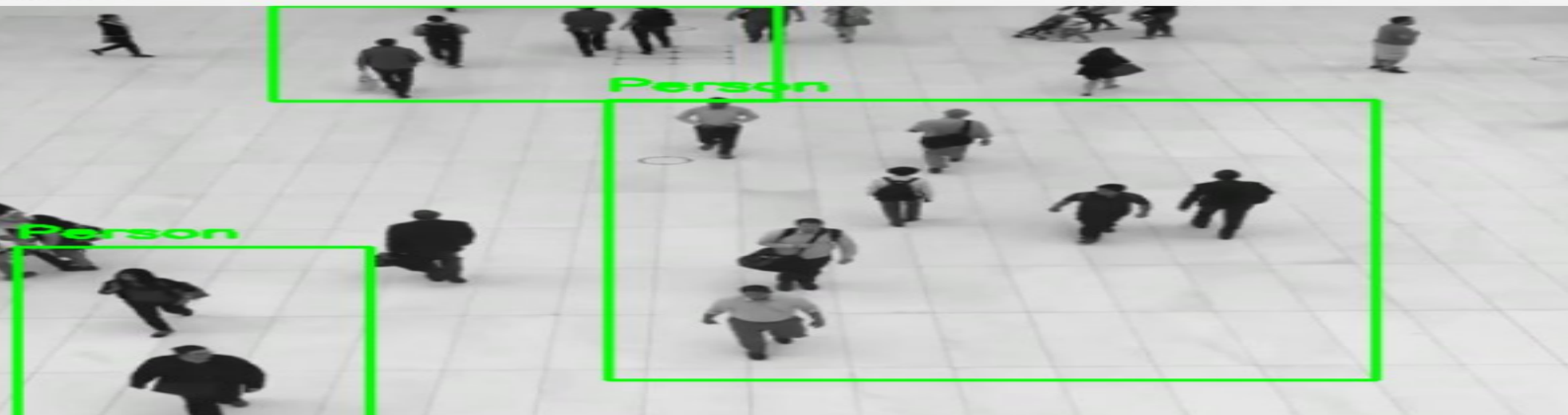


Fig 1 — STATUS: NORMAL — Green bounding boxes, no alert triggered

STATUS: ABNORMAL

Violent incident detected. Red bounding boxes activated, full-width red warning banner displayed at top of frame, and audio buzzer fired simultaneously on a background thread.

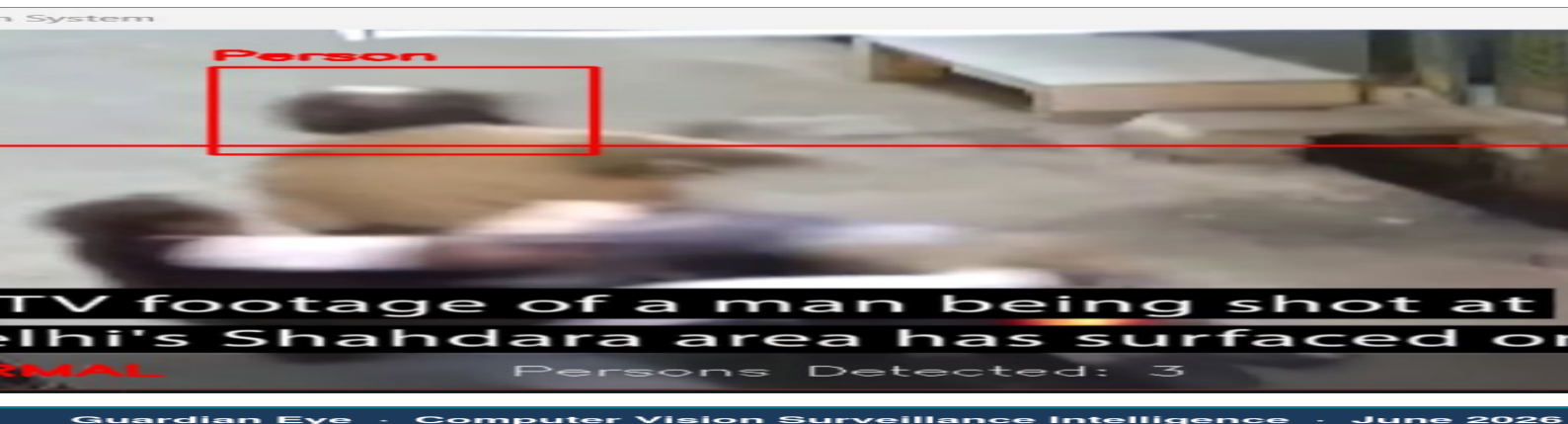


Fig 2 — STATUS: ABNORMAL — Red bounding boxes, warning banner, audio alert